

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Master of Technology (Heat Power Engineering) SEMESTER - 3 Winter 2025 (Regular)

Course :Master of Technology (Heat Power Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 24AF2914OE301A - PROJECT MANAGEMENT

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

- 1.All the questions are compulsory.
 - 2.Use of non-programmable scientific calculators is allowed.
 - 3.Assume suitable data wherever necessary and mention it clearly.
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1 Solve Any Two of the following.

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|-------|---|---|
| Q1(a) | Explain the concept of Project Management. Discuss its definition, objectives, and importance in modern engineering and industrial organizations. | 6 |
| Q1(b) | Discuss the concept of Project Organization. Explain different types of project organizational structures and their advantages and limitations. | 6 |
| Q1(c) | Explain various Time Estimation Methods used in project management. Discuss their advantages and limitations. | 6 |

2 Solve Any Two of the following.

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|-------|---|---|
| Q2(a) | Describe the methods of Project Cost Estimation and Budgeting. Explain how accurate cost estimation contributes to project success. | 6 |
| Q2(b) | Describe the process of Developing a Project Plan. Explain how scope, schedule, cost, and resources are integrated into a comprehensive project plan. | 6 |
| Q2(c) | What is Project Cash Flow Analysis? Explain its objectives and significance in project planning and financial control. | 6 |

3 Solve Any Two of the following.

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|-------|---|---|
| Q3(a) | Discuss the concept of Project Scheduling with Resource Constraints. Why are resource constraints critical in real-life project management? | 6 |
| Q3(b) | What is Resource Allocation? Explain the procedure for allocating limited resources among competing project activities. | 6 |
| Q3(c) | Explain the concept of Time–Cost Trade-off in project management. Why is it necessary to balance time and cost in projects? | 6 |

4 Solve Any Two of the following.

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|-------|--|---|
| Q4(a) | Explain the concept of Project Implementation. Discuss the key activities involved during the implementation phase of a project. | 6 |
| Q4(b) | Explain PERT–Cost Control techniques. How are time, cost, and performance integrated in PERT-based project monitoring? | 6 |
| Q4(c) | What is Contract Management? Explain the stages involved in contract management during the life cycle of a project. | 6 |

5 Solve Any Two of the following.

- Q5(a) Describe different types of contracts used in project management. Discuss their advantages, limitations, and suitability for different projects. 6
- Q5(b) What is meant by Post-Project Analysis? Explain its objectives and importance in improving future project performance. 6
- Q5(c) Describe the methods and techniques used in Post-Project Evaluation. Explain how lessons learned contribute to organizational knowledge. 6

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